

LAB # 9 : Joint Distribution, Correlation, and Regression

Introduction

This lab is concerned with joint probability distributions and correlation. We also study the regression lines.

Joint Distribution

We can compute probabilities, expected values, and the correlation coefficient for joint distributions (both discrete and continuous). We illustrate by taking examples.

Example 1: If the joint probability distribution of X and Y is given by

$$f(x, y) = \frac{x + y}{30}, \text{ for } x = 0, 1, 2, 3; y = 0, 1, 2.$$

- (a) Find the values of the joint distribution in matrix form.
- (b) Verify that it is indeed a pmf.
- (c) Find the marginal distributions $g(x)$ and $h(y)$.
- (d) Find conditional probability $P(X = 2 \mid Y = 1)$.
- (e) Find $E(X)$, $E(Y)$, $Var(X)$, and $Var(Y)$.
- (f) Find $E(XY)$, $Cov(X, Y)$, and correlation coefficient.

Sol.

```
(a)  x_vals <- 0:3
     y_vals <- 0:2
     joint_pmf <- outer(x_vals, y_vals, function(x, y) (x + y) / 30)
     print(joint_pmf)
           [,1]      [,2]      [,3]
[1,] 0.00000000 0.03333333 0.06666667
[2,] 0.03333333 0.06666667 0.10000000
[3,] 0.06666667 0.10000000 0.13333333
[4,] 0.10000000 0.13333333 0.16666667

(b)  sum(joint_pmf)
[1] 1

(c)  # Marginal distribution
     g_x <- rowSums(joint_pmf)
     print(g_x)
[1] 0.1 0.2 0.3 0.4
     h_y <- colSums(joint_pmf)
     print(h_y)
[1] 0.2000000 0.3333333 0.4666667

(d)  condprob <- joint_pmf[3,2]/h_y[2] # x=2,y=1
     condprob
[1] 0.3
```

```

(e)   E_x <- sum(x_vals*g_x)
      E_x
      [1] 2
      E_y <- sum(y_vals*h_y)
      E_y
      [1] 1.266667
      E_x2 <- sum(x_vals^2*g_x)
      E_x2
      [1] 5
      E_y2 <- sum(y_vals^2*h_y)
      E_y2
      [1] 2.2
      var_x <- E_x2-E_x^2
      var_x
      [1] 1
      var_y <- E_y2-E_y^2
      var_y
      [1] 0.5955556
(f)   joint_xy <- outer(x_vals, y_vals, function(x, y) x*y*(x + y) / 30)
      print(joint_xy)
           [,1]      [,2]      [,3]
[1,]      0      0.00000000      0.0000000
[2,]      0      0.06666667      0.2000000
[3,]      0      0.20000000      0.5333333
[4,]      0      0.40000000      1.0000000
      E_xy <- sum(joint_xy)
      print(E_xy)
      [1] 2.4
      cov <- E_xy-E_x*E_y
      cov
      [1] -0.1333333
      r_xy <- cov/sqrt(var_x*var_y)
      r_xy
      [1] -0.1727737

```

Example 2: If the joint probability distribution of X and Y is given by

$$f(x, y) = \frac{2}{5}(2x + 3y), \quad 0 \leq x \leq 1, 0 \leq y \leq 1.$$

- (a) Verify that it is indeed a pdf.
- (b) Find the marginal distributions $g(1)$ and $h(0)$.
- (c) Find $E(XY)$.

Sol.

```

(a)   install.packages("pracma")
      library(pracma)
      f <- function(x, y) (2/5) * (2*x+3*y)
      I <- integral2(f, xmin=0, xmax=1, ymin=0, ymax=1)
      print(I) 1 error
      [1] 1.249001e-16
      print(I$Q)
      [1] 1
(b)   gx.1 <- function(y) f(1, y)
      gx <- integral(gx.1, 0, 1)
      print(gx)
      [1] 1.4

```

```

hy_0<- function(x) f(x,0)
hy <-integral(hy_0,0,1)
print(hy)
[1] 0.4
(c) f2 <- function(x,y) x*y*f(x,y)
I2 <- integral2(f2,0,1,0,1)
print(I2$Q)
[1] 0.3333333

```

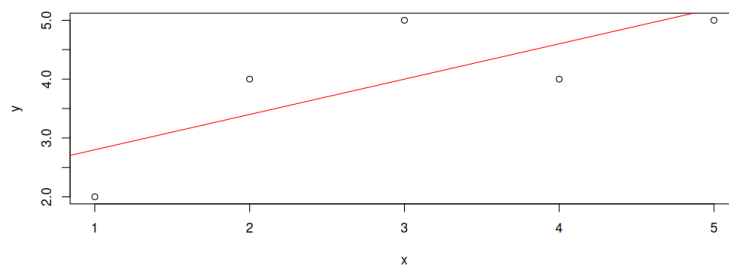
Correlation and Regression

We can find the correlation coefficient and regression line in *R*.

```

x <- c(1, 2, 3, 4, 5)
y <- c(2, 4, 5, 4, 5)
cor(x,y)
[1] 0.7745967
plot(x,y)
# Fit the linear model
model <- lm(y ~ x)
# View the summary of the model
summary(model)
#Add the regression line to the scatter diagram
abline(model, col = "red")

```



To illustrate more about correlations, we will be using the **mtcars** data set that is built into *R*. For simplicity's sake, we will only be looking at three of the variables; we will therefore need to subset the data. First, we will load the full mtcars data set into a dataframe, and look at what the variables are. We can see that there are 11 variables, some of which are not continuous. We will keep mpg (miles per gallon), hp (horse power), and wt (weight). To do this, we make a new dataframe (data1).

```

data <- mtcars
# Look at the included variables
head(data)
# Output

```

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
Mazda RX4	21.0	6	160	110	3.90	2.620	16.46	0	1	4	4
Mazda RX4 Wag	21.0	6	160	110	3.90	2.875	17.02	0	1	4	4
Datsun 710	22.8	4	108	93	3.85	2.320	18.61	1	1	4	1
Hornet 4 Drive	21.4	6	258	110	3.08	3.215	19.44	1	0	3	1
Hornet Sportabout	18.7	8	360	175	3.15	3.440	17.02	0	0	3	2

```

Valiant          18.1    6  225  105  2.76  3.460  20.22  1  0    3    1
#Load tidyverse package
library(tidyverse)
#Only keep 3 variables: mpg, hp, and wt
library(dplyr) # to use pipe %>%
data1 <- data %>%
  select(c(mpg, hp, wt))
      mpg  hp    wt
Mazda RX4      21.0 110 2.620
Mazda RX4 Wag  21.0 110 2.875
Datsun 710      22.8  93 2.320
Hornet 4 Drive  21.4 110 3.215
Hornet Sportabout 18.7 175 3.440
Valiant        18.1 105 3.46
library(ggplot2)
#mpg-hp scatterplot
ggplot(data1, aes(x = mpg, y = hp)) +
  geom_point()
#hp-wt scatterplot
ggplot(data1, aes(x = hp, y = wt)) +
  geom_point()
#mpg-wt scatterplot
ggplot(data1, aes(x = mpg, y = wt)) +
  geom_point()
#Creating multiple scatterplots
pairs(data1[, c("mpg", "hp", "wt")])
#Run the test
cor.test(data1$mpg, data1$hp)

```

Exercises:

1. In a communication system, let X be the number of packets lost $(0, 1, 2)$ and Y be the number of packets retransmitted $(0, 1, 2, 3)$. The joint pmf is given by

$$f(x, y) = k(x + 2y + 1), \quad x = 0, 1, 2; y = 0, 1, 2, 3.$$

- (a) Find the value of k .
 - (b) Construct the joint probability matrix in R and find the marginal distributions of X and Y .
 - (c) Compute $E(X)$, $E(Y)$, $Var(X)$, $Var(Y)$, $Cov(X, Y)$, and the correlation coefficient.
 - (d) Are X and Y independent? Justify.
 - (e) Find $P(X \leq 1 \mid Y = 2)$.
2. Let X be the normalized CPU load $(0 \leq x \leq 1)$. Let Y be the normalized temperature $(0 \leq y \leq 2)$. The joint pdf is given as

$$f(x, y) = kxy, \quad 0 \leq x \leq 1, 0 \leq y \leq 2.$$

- (a) Find k .
 - (b) Find the marginal pdfs $g(x)$ and $h(y)$.
 - (c) Compute $E(X)$, $E(Y)$, and $E(XY)$ and $Cov(X, Y)$.
 - (d) Are X and Y independent?
 - (e) Find $P(X > 0.5, Y > 1)$.
3. A student records the following data:

Input Size (MB)	10	20	30	40	50	60
Runtime (ms)	15	28	41	55	68	82

- (a) Plot the scatter diagram.
 - (b) Compute the correlation coefficient.
 - (c) Fit the regression line $\hat{y} = a + bx$.
 - (d) Predict the runtime for a 75 MB input.
4. The following table shows the amount of rainfall (in mm) and wheat production (in tonnes) for eight years:

Rainfall	60	62	65	71	73	76	84	82
Wheat Production	37	38	41	42	45	48	52	54

Plot the scatter diagram and fit a regression line to predict wheat production from rainfall. Further, estimate the wheat production corresponding to 68 mm of rainfall.